

# Lab Activity

Visualization Evaluation: Python + R, Code & Output, 5 Activities

Each activity: Python code + output, followed by R code + output

Date: 20/07/2026

## Activity 1

# Hospital Patient Admission Analysis

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## PYTHON

### Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
import squarify

sns.set_style("whitegrid")
np.random.seed(1)

departments = ['Cardiology', 'Neurology', 'Pediatrics', 'Orthopedics', 'Emergency']
months = pd.date_range("2025-01-01", periods=12, freq='MS').strftime('%b')
base_load = {'Cardiology': 60, 'Neurology': 40, 'Pediatrics': 50, 'Orthopedics': 45, 'Emergency': 90}

rows = []
for dept in departments:
    for m in months:
        admissions = max(5, int(np.random.normal(base_load[dept], base_load[dept] * 0.2)))
        rows.append([dept, m, admissions])
df = pd.DataFrame(rows, columns=['Department', 'Month', 'Admissions'])

annual_totals = df.groupby('Department')['Admissions'].sum().sort_values(ascending=False)

plt.figure(figsize=(7, 5))
annual_totals.plot(kind='bar', color='steelblue')
plt.title("Total Annual Admissions by Department (Bar Chart)")
plt.xlabel("Department"); plt.ylabel("Total Admissions")
plt.tight_layout()
plt.show()

pivot = df.pivot(index='Month', columns='Department', values='Admissions').reindex(months)
pivot.plot(kind='bar', stacked=True, figsize=(9, 5), colormap='tab10')
plt.title("Monthly Admissions by Department (Stacked Bar Chart)")
plt.xlabel("Month"); plt.ylabel("Admissions")
plt.legend(title="Department", bbox_to_anchor=(1.02, 1), loc='upper left')
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
squarify.plot(sizes=annual_totals.values, label=[f"{d}\n{v}" for d, v in annual_totals.items()],
              color=sns.color_palette("Set2", len(annual_totals)))
plt.title("Share of Total Admissions by Department (Treemap)")
plt.axis('off')
plt.tight_layout()
plt.show()

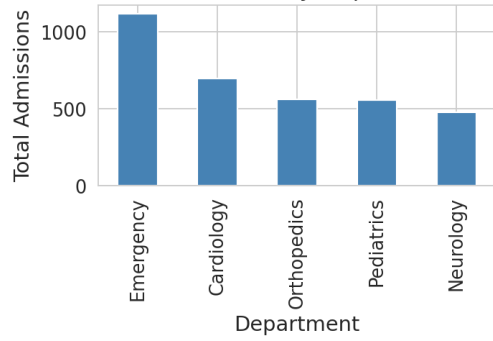
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Department', y='Admissions', hue='Department', palette='Set3', legend=False)
plt.title("Monthly Admission Variability by Department (Box Plot)")
plt.xlabel("Department"); plt.ylabel("Monthly Admissions")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
plt.hist(df['Admissions'], bins=15, color='mediumpurple', edgecolor='black')
plt.title("Distribution of Monthly Admission Counts (Histogram)")
plt.xlabel("Admissions per Month"); plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

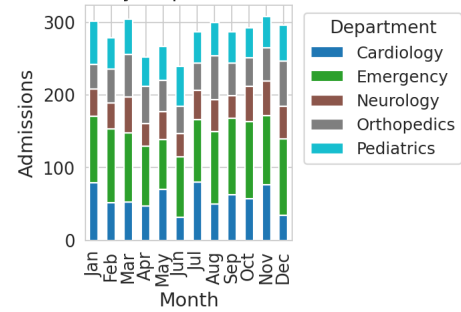
print("Evaluation: see accompanying write-up for visualization justification.")
```

### Output

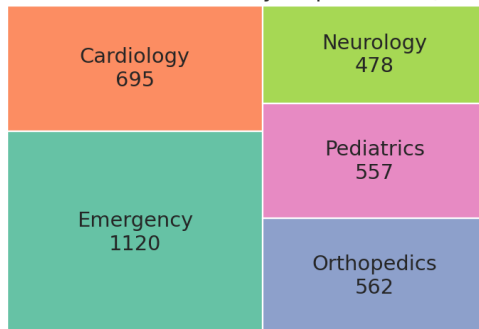
Total Annual Admissions by Department (Bar Chart)



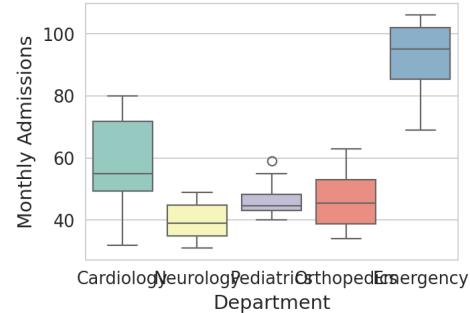
Monthly Admissions by Department (Stacked Bar Chart)



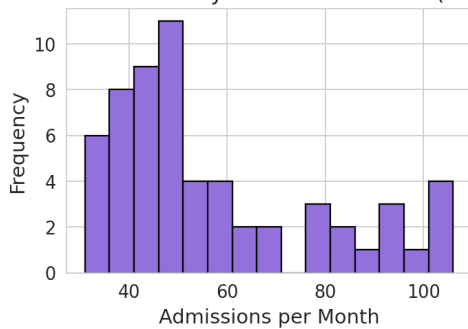
Share of Total Admissions by Department (Treemap)



Monthly Admission Variability by Department (Box Plot)



Distribution of Monthly Admission Counts (Histogram)



## R

### Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

departments <- c('Cardiology','Neurology','Pediatrics','Orthopedics','Emergency')
months <- month.abb
base_load <- c(Cardiology=60, Neurology=40, Pediatrics=50, Orthopedics=45, Emergency=90)

df <- data.frame()
for (dept in departments) {
  admissions <- pmax(5, round(rnorm(12, base_load[[dept]], base_load[[dept]]*0.2)))
  df <- rbind(df, data.frame(Department=dept, Month=months, Admissions=admissions))
}
df$Month <- factor(df$Month, levels=months)

annual_totals <- df %>% group_by(Department) %>% summarise(Total=sum(Admissions)) %>%
  arrange(desc(Total))
annual_totals$Department <- factor(annual_totals$Department, levels=annual_totals$Department)

p1 <- ggplot(annual_totals, aes(x=Department, y=Total, fill=Department)) +
  geom_col(show.legend=FALSE) +
  labs(title="Total Annual Admissions by Department (Bar Chart)", y="Total Admissions") +
  theme_minimal()
print(p1)
p2 <- ggplot(df, aes(x=Month, y=Admissions, fill=Department)) +
```

```

geom_col() +
  labs(title="Monthly Admissions by Department (Stacked Bar Chart)", y="Admissions") +
  theme_minimal()
print(p2)

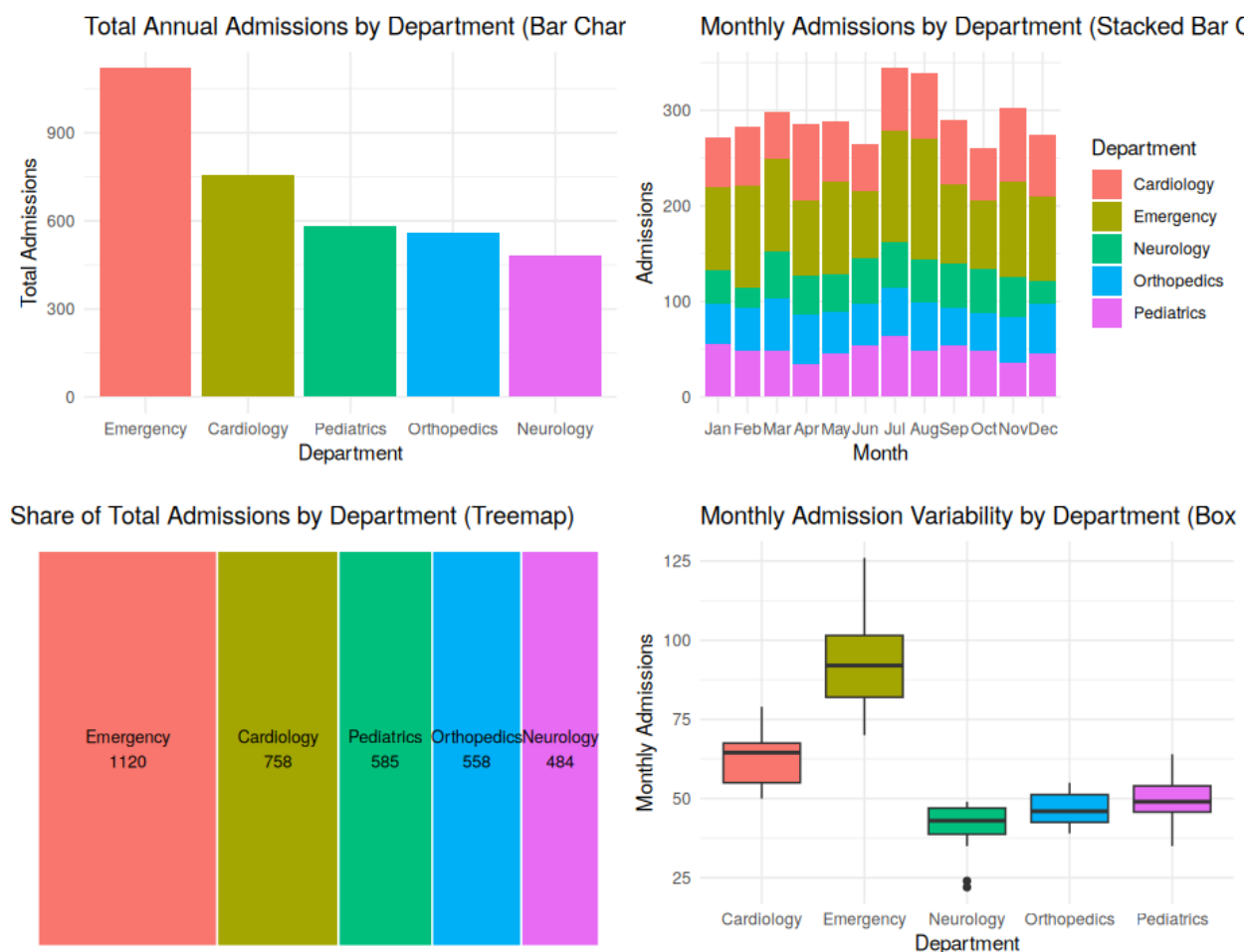
tm <- annual_totals %>% arrange(desc(Total)) %>%
  mutate(frac = Total / sum(Total), xmax = cumsum(frac), xmin = lag(xmax, default=0))
p3 <- ggplot(tm) +
  geom_rect(aes(xmin=xmin, xmax=xmax, ymin=0, ymax=1, fill=Department), color="white") +
  geom_text(aes(x=(xmin+xmax)/2, y=0.5, label=paste0(Department, "\n", Total)), size=3.2) +
  labs(title="Share of Total Admissions by Department (Treemap)") +
  theme_void() + theme(legend.position="none")
print(p3)

p4 <- ggplot(df, aes(x=Department, y=Admissions, fill=Department)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Monthly Admission Variability by Department (Box Plot)", y="Monthly Admissions") +
  theme_minimal()
print(p4)

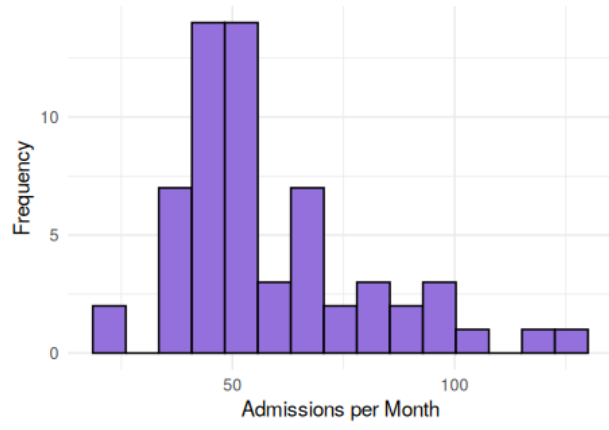
p5 <- ggplot(df, aes(x=Admissions)) +
  geom_histogram(bins=15, fill="mediumpurple", color="black") +
  labs(title="Distribution of Monthly Admission Counts (Histogram)",
        x="Admissions per Month", y="Frequency") +
  theme_minimal()
print(p5)

```

## Output



Distribution of Monthly Admission Counts (Histogram)



## Activity 2

# E-Commerce Customer Spending Behaviour

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## PYTHON

### Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

n = 300
categories = ['Electronics', 'Fashion', 'Grocery', 'Home', 'Beauty']
df = pd.DataFrame({
    'CustomerID': range(1, n + 1),
    'SpendAmount': np.random.lognormal(mean=4.2, sigma=0.9, size=n).round(2),
    'OrderFrequency': np.random.poisson(6, n),
    'Category': np.random.choice(categories, n)
})

plt.figure(figsize=(7, 5))
plt.hist(df['SpendAmount'], bins=30, color='steelblue', edgecolor='black')
plt.title("Distribution of Customer Spend Amount (Histogram)")
plt.xlabel("Spend Amount ($)"); plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.violinplot(data=df, x='Category', y='SpendAmount', hue='Category', palette='Set2', legend=False)
plt.title("Spend Amount Distribution by Category (Violin Plot)")
plt.xlabel("Category"); plt.ylabel("Spend Amount ($)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Category', y='SpendAmount', hue='Category', palette='Set3', legend=False)
plt.title("Spend Amount Outliers by Category (Box Plot)")
plt.xlabel("Category"); plt.ylabel("Spend Amount ($)")
plt.tight_layout()
plt.show()

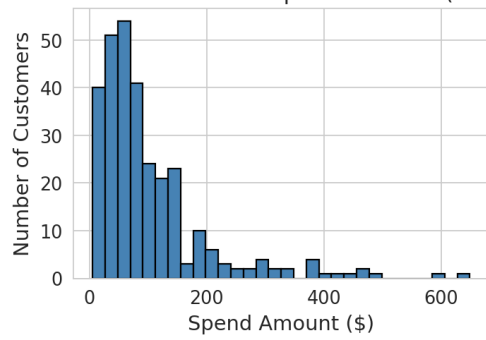
plt.figure(figsize=(7, 5))
sns.kdeplot(df['SpendAmount'], fill=True, color='darkorange')
plt.title("Spend Amount Density (Density Plot)")
plt.xlabel("Spend Amount ($)"); plt.ylabel("Density")
plt.tight_layout()
plt.show()

avg_spend = df.groupby('Category')['SpendAmount'].mean().sort_values(ascending=False)
plt.figure(figsize=(7, 5))
avg_spend.plot(kind='bar', color='mediumseagreen')
plt.title("Average Spend Amount by Category (Bar Chart)")
plt.xlabel("Category"); plt.ylabel("Average Spend ($)")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

### Output

Distribution of Customer Spend Amount (Histogram)



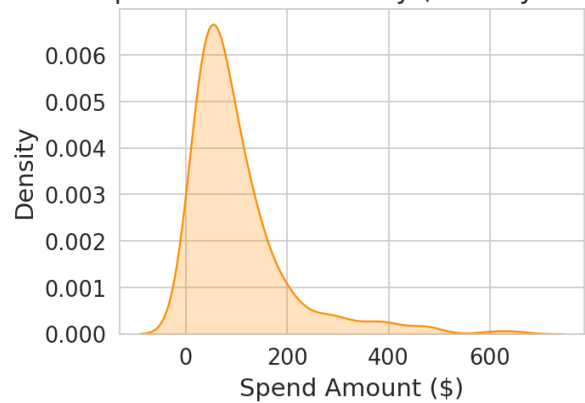
Spend Amount Distribution by Category (Violin Plot)



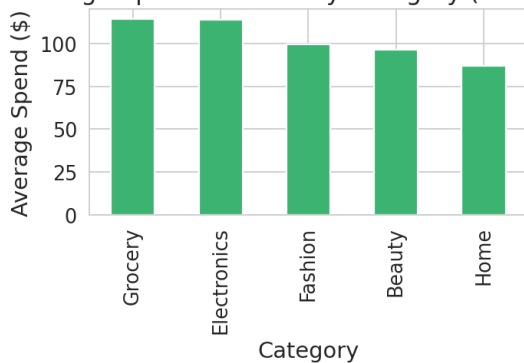
Spend Amount Outliers by Category (Box Plot)



Spend Amount Density (Density Plot)



Average Spend Amount by Category (Bar Chart)



## R

### Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

n <- 300
categories <- c('Electronics', 'Fashion', 'Grocery', 'Home', 'Beauty')
df <- data.frame(
  CustomerID = 1:n,
  SpendAmount = round(rlnorm(n, meanlog=4.2, sdlog=0.9), 2),
  OrderFrequency = rpois(n, 6),
  Category = sample(categories, n, replace=TRUE)
)

p1 <- ggplot(df, aes(x=SpendAmount)) +
  geom_histogram(bins=30, fill="steelblue", color="black") +
  labs(title="Distribution of Customer Spend Amount (Histogram)",
       x="Spend Amount ($)", y="Number of Customers") +
  theme_minimal()
print(p1)
p2 <- ggplot(df, aes(x=Category, y=SpendAmount, fill=Category)) +
```

```

geom_violin(show.legend=FALSE) +
labs(title="Spend Amount Distribution by Category (Violin Plot)", y="Spend Amount ($)") +
theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=Category, y=SpendAmount, fill=Category)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Spend Amount Outliers by Category (Box Plot)", y="Spend Amount ($)") +
  theme_minimal()
print(p3)

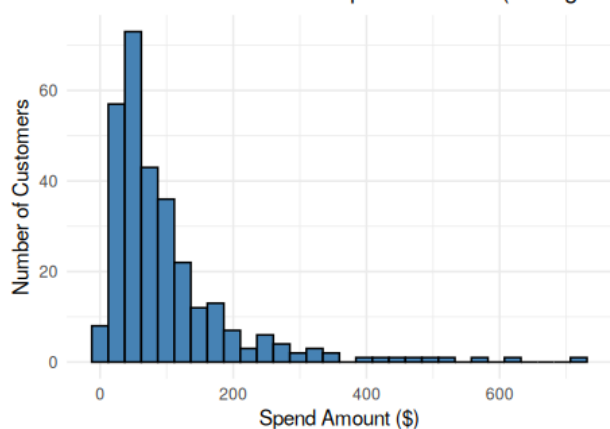
p4 <- ggplot(df, aes(x=SpendAmount)) +
  geom_density(fill="darkorange", alpha=0.6) +
  labs(title="Spend Amount Density (Density Plot)", x="Spend Amount ($)", y="Density") +
  theme_minimal()
print(p4)

avg_spend <- df %>% group_by(Category) %>% summarise(AvgSpend=mean(SpendAmount)) %>%
  arrange(desc(AvgSpend))
avg_spend$Category <- factor(avg_spend$Category, levels=avg_spend$Category)
p5 <- ggplot(avg_spend, aes(x=Category, y=AvgSpend)) +
  geom_col(fill="mediumseagreen") +
  labs(title="Average Spend Amount by Category (Bar Chart)", y="Average Spend ($)") +
  theme_minimal()
print(p5)

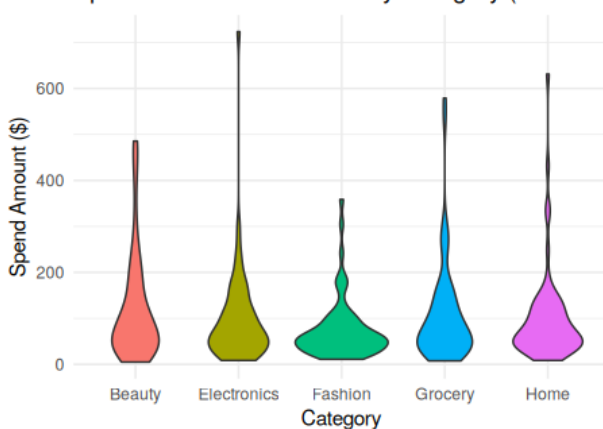
```

## Output

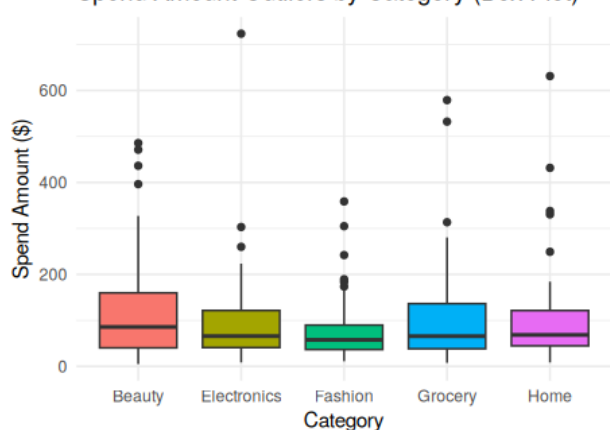
Distribution of Customer Spend Amount (Histogram)



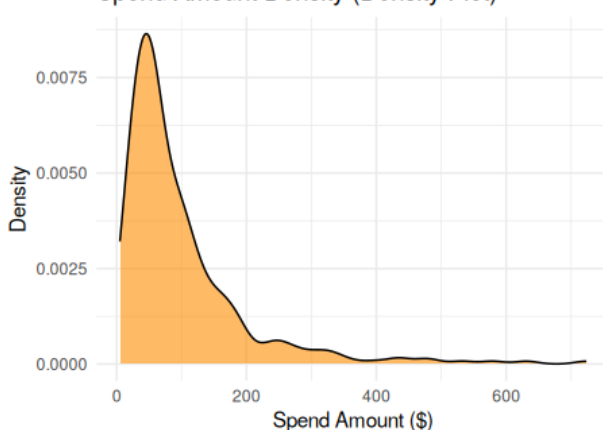
Spend Amount Distribution by Category (Violin Plot)

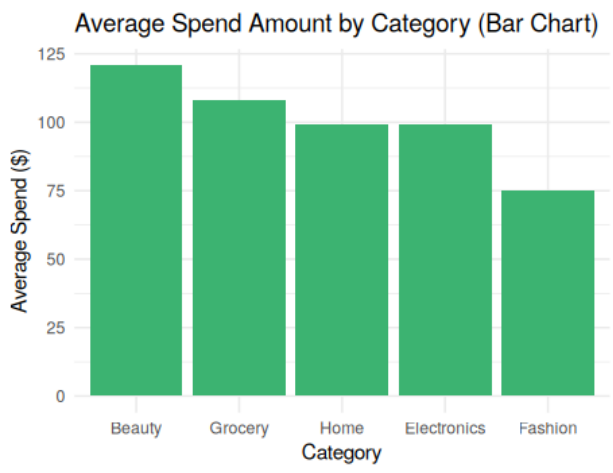


Spend Amount Outliers by Category (Box Plot)



Spend Amount Density (Density Plot)





## Activity 3

# University Examination Performance Dashboard

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## PYTHON

### Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

departments = ['CSE', 'ECE', 'Mechanical', 'Civil', 'IT']
n_per_dept = 60
rows = []
dept_mean = {'CSE': 78, 'ECE': 72, 'Mechanical': 65, 'Civil': 68, 'IT': 80}
classes = ['Section A', 'Section B', 'Section C']

for dept in departments:
    for i in range(n_per_dept):
        marks = np.clip(np.random.normal(dept_mean[dept], 12), 0, 100)
        cls = np.random.choice(classes)
        rows.append([dept, cls, round(marks, 1)])
df = pd.DataFrame(rows, columns=['Department', 'Class', 'Marks'])

avg_marks = df.groupby('Department')['Marks'].mean().sort_values(ascending=False)
plt.figure(figsize=(7, 5))
avg_marks.plot(kind='bar', color='steelblue')
plt.title("Average Marks by Department (Bar Chart)")
plt.xlabel("Department"); plt.ylabel("Average Marks")
plt.tight_layout()
plt.show()

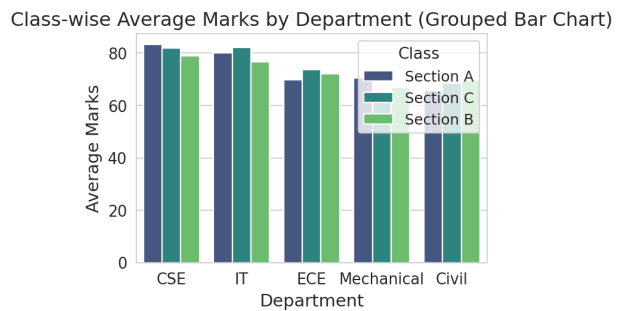
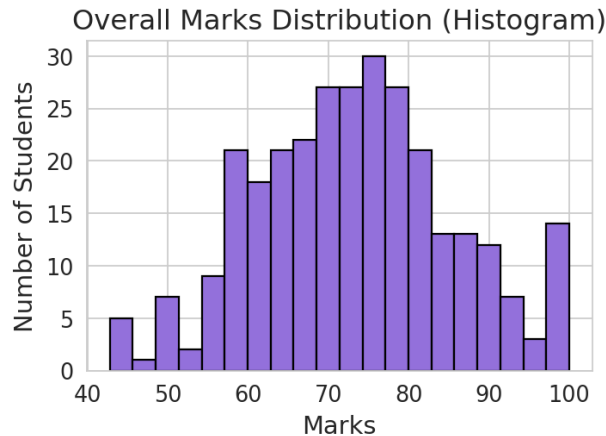
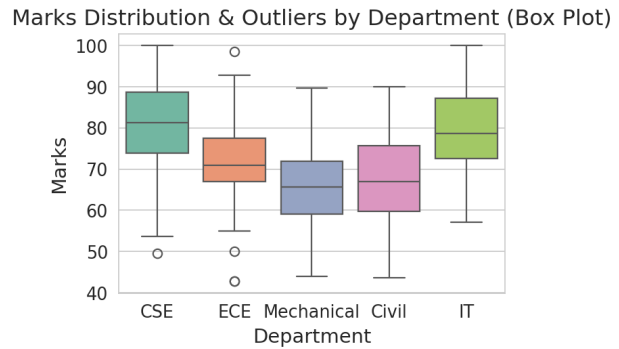
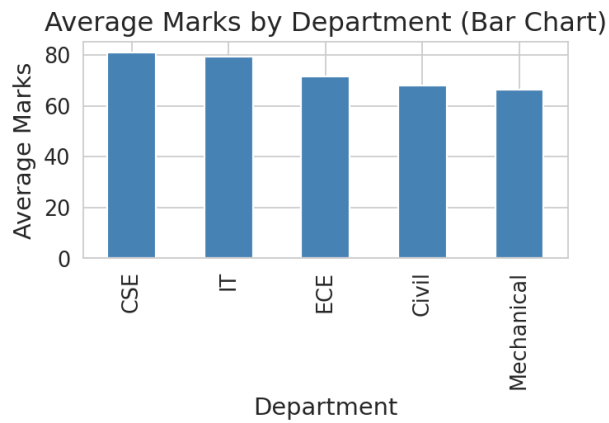
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Department', y='Marks', hue='Department', palette='Set2', legend=False)
plt.title("Marks Distribution & Outliers by Department (Box Plot)")
plt.xlabel("Department"); plt.ylabel("Marks")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
plt.hist(df['Marks'], bins=20, color='mediumpurple', edgecolor='black')
plt.title("Overall Marks Distribution (Histogram)")
plt.xlabel("Marks"); plt.ylabel("Number of Students")
plt.tight_layout()
plt.show()

class_avg = df.groupby(['Department', 'Class', observed=True])['Marks'].mean().reset_index()
top_classes = class_avg.sort_values('Marks', ascending=False)
plt.figure(figsize=(9, 5))
sns.barplot(data=top_classes, x='Department', y='Marks', hue='Class', palette='viridis')
plt.title("Class-wise Average Marks by Department (Grouped Bar Chart)")
plt.xlabel("Department"); plt.ylabel("Average Marks")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

### Output



## R

### Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

departments <- c('CSE','ECE','Mechanical','Civil','IT')
n_per_dept <- 60
dept_mean <- c(CSE=78, ECE=72, Mechanical=65, Civil=68, IT=80)
classes <- c('Section A','Section B','Section C')

df <- data.frame()
for (dept in departments) {
  marks <- pmin(pmax(rnorm(n_per_dept, dept_mean[[dept]], 12), 0), 100)
  cls <- sample(classes, n_per_dept, replace=TRUE)
  df <- rbind(df, data.frame(Department=dept, Class=cls, Marks=round(marks,1)))
}

avg_marks <- df %>% group_by(Department) %>% summarise(AvgMarks=mean(Marks)) %>%
  arrange(desc(AvgMarks))
avg_marks$Department <- factor(avg_marks$Department, levels=avg_marks$Department)
p1 <- ggplot(avg_marks, aes(x=Department, y=AvgMarks)) +
  geom_col(fill="steelblue") +
  labs(title="Average Marks by Department (Bar Chart)", y="Average Marks") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x=Department, y=Marks, fill=Department)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Marks Distribution & Outliers by Department (Box Plot)", y="Marks") +
  theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=Marks)) +
  geom_histogram(bins=20, fill="mediumpurple", color="black") +
  labs(title="Overall Marks Distribution (Histogram)", x="Marks", y="Number of Students") +
  theme_minimal()
print(p3)
```

```
class_avg <- df %>% group_by(Department, Class) %>% summarise(AvgMarks=mean(Marks), .groups='drop')
p4 <- ggplot(class_avg, aes(x=Department, y=AvgMarks, fill=Class)) +
  geom_col(position=position_dodge()) +
  labs(title="Class-wise Average Marks by Department (Grouped Bar Chart)", y="Average Marks") +
  theme_minimal()
print(p4)
```

## Output



## Activity 4

# Climate and Rainfall Distribution Study

---

## PYTHON

### Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

cities = ['Chennai', 'Mumbai', 'Delhi', 'Bengaluru']
months = pd.date_range("2025-01-01", periods=12, freq='MS').strftime('%b')
monsoon_months = ['Jun', 'Jul', 'Aug', 'Sep']

rows = []
for city in cities:
    for m in months:
        base_rain = 180 if m in monsoon_months else 30
        rainfall = max(0, np.random.normal(base_rain, base_rain * 0.3))
        temp = np.random.normal(28, 4)
        humidity = np.random.normal(65, 10)
        rows.append([city, m, round(rainfall, 1), round(temp, 1), round(humidity, 1)])
df = pd.DataFrame(rows, columns=['City', 'Month', 'Rainfall', 'Temperature', 'Humidity'])

plt.figure(figsize=(7, 5))
plt.hist(df['Rainfall'], bins=20, color='steelblue', edgecolor='black')
plt.title("Distribution of Monthly Rainfall (Histogram)")
plt.xlabel("Rainfall (mm)"); plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

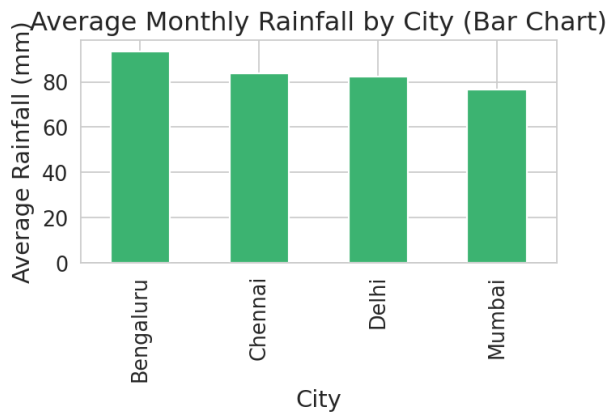
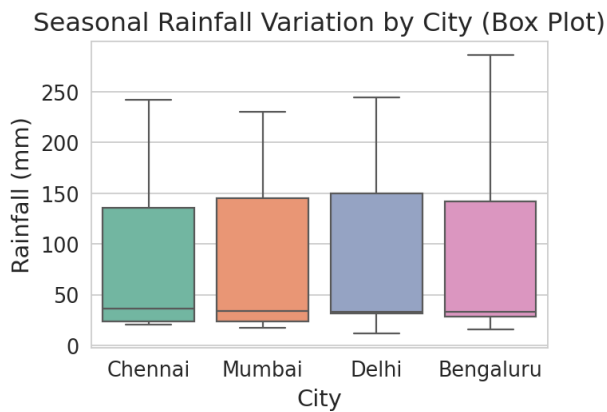
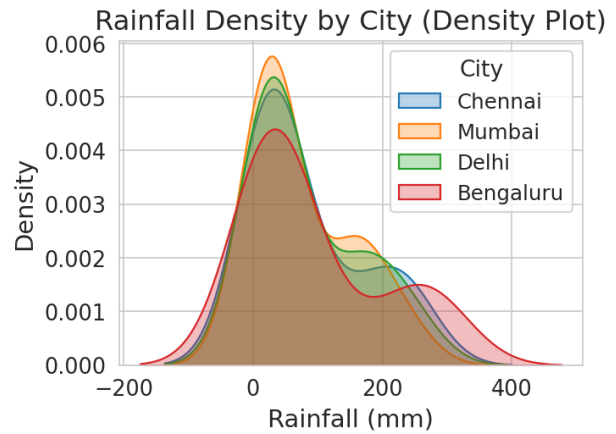
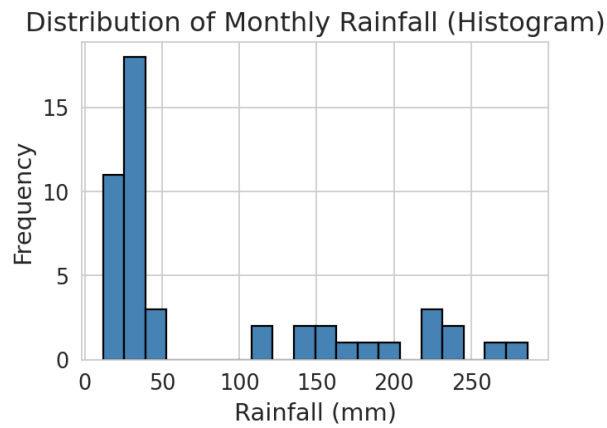
plt.figure(figsize=(8, 5))
for city in cities:
    sns.kdeplot(df[df.City == city]['Rainfall'], label=city, fill=True, alpha=0.3)
plt.title("Rainfall Density by City (Density Plot)")
plt.xlabel("Rainfall (mm)"); plt.ylabel("Density")
plt.legend(title="City")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='City', y='Rainfall', hue='City', palette='Set2', legend=False)
plt.title("Seasonal Rainfall Variation by City (Box Plot)")
plt.xlabel("City"); plt.ylabel("Rainfall (mm)")
plt.tight_layout()
plt.show()

avg_rain = df.groupby('City')['Rainfall'].mean().sort_values(ascending=False)
plt.figure(figsize=(7, 5))
avg_rain.plot(kind='bar', color='mediumseagreen')
plt.title("Average Monthly Rainfall by City (Bar Chart)")
plt.xlabel("City"); plt.ylabel("Average Rainfall (mm)")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

### Output



## R

### Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

cities <- c('Chennai', 'Mumbai', 'Delhi', 'Bengaluru')
months <- month.abb
monsoon_months <- c('Jun', 'Jul', 'Aug', 'Sep')

df <- data.frame()
for (city in cities) {
  for (m in months) {
    base_rain <- if (m %in% monsoon_months) 180 else 30
    rainfall <- max(0, rnorm(1, base_rain, base_rain*0.3))
    temp <- rnorm(1, 28, 4)
    humidity <- rnorm(1, 65, 10)
    df <- rbind(df, data.frame(City=city, Month=m, Rainfall=round(rainfall,1),
                             Temperature=round(temp,1), Humidity=round(humidity,1)))
  }
}

p1 <- ggplot(df, aes(x=Rainfall)) +
  geom_histogram(bins=20, fill="steelblue", color="black") +
  labs(title="Distribution of Monthly Rainfall (Histogram)", x="Rainfall (mm)", y="Frequency") +
  theme_minimal()
print(p1)

p2 <- ggplot(df, aes(x=Rainfall, fill=City)) +
  geom_density(alpha=0.3) +
  labs(title="Rainfall Density by City (Density Plot)", x="Rainfall (mm)", y="Density") +
  theme_minimal()
print(p2)

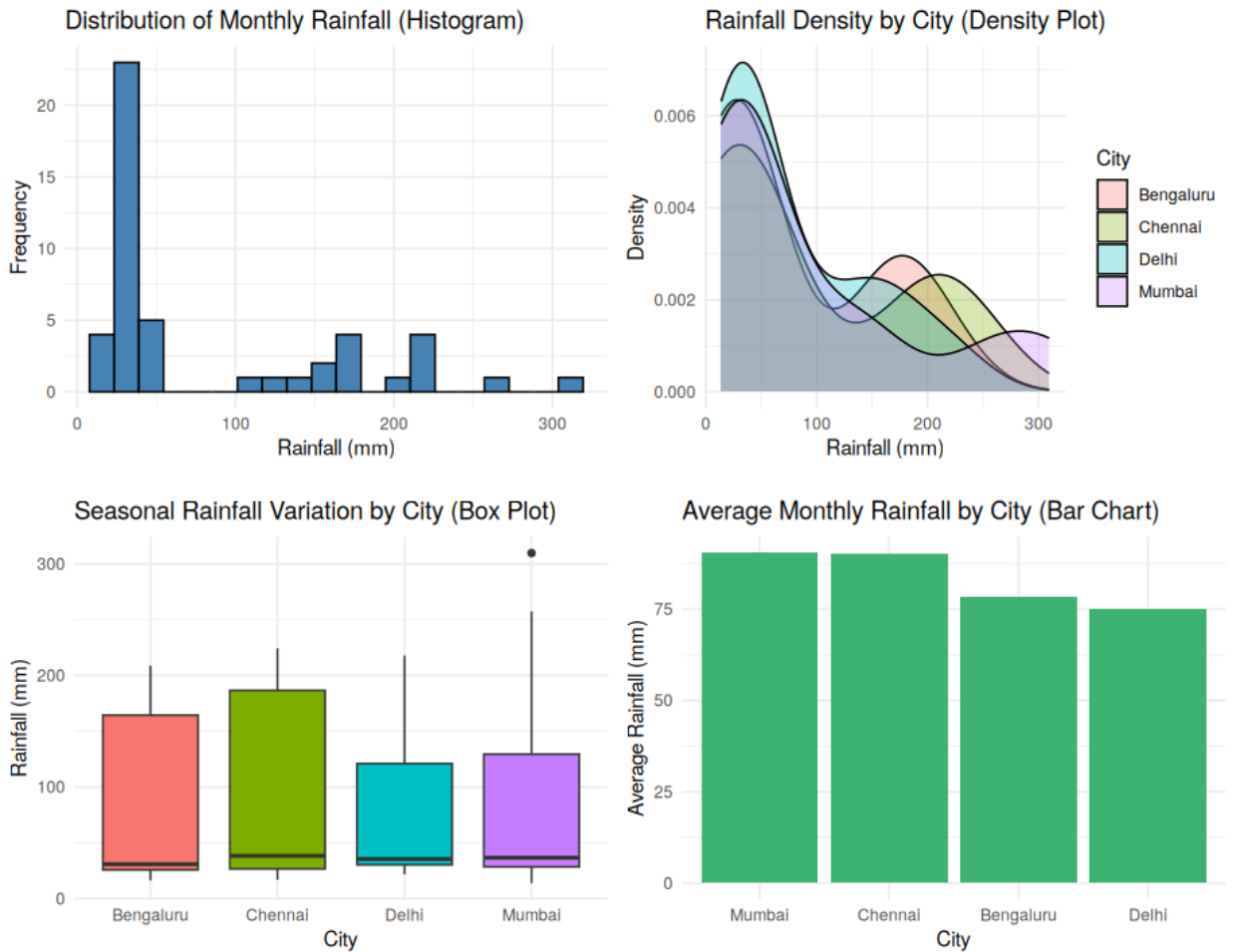
p3 <- ggplot(df, aes(x=City, y=Rainfall, fill=City)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Seasonal Rainfall Variation by City (Box Plot)", y="Rainfall (mm)") +
  theme_minimal()
print(p3)
```

```

avg_rain <- df %>% group_by(City) %>% summarise(AvgRainfall=mean(Rainfall)) %>%
  arrange(desc(AvgRainfall))
avg_rain$City <- factor(avg_rain$City, levels=avg_rain$City)
p4 <- ggplot(avg_rain, aes(x=City, y=AvgRainfall)) +
  geom_col(fill="mediumseagreen") +
  labs(title="Average Monthly Rainfall by City (Bar Chart)", y="Average Rainfall (mm)") +
  theme_minimal()
print(p4)

```

## Output



## Activity 5

# Bank Loan Risk Assessment

---

## PYTHON

### Code

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np

sns.set_style("whitegrid")
np.random.seed(1)

n = 300
df = pd.DataFrame({
    'CustomerID': range(1, n + 1),
    'LoanAmount': np.random.gamma(shape=3, scale=8000, size=n).round(0),
    'Income': np.random.normal(55000, 18000, n).clip(15000).round(0),
    'CreditScore': np.random.normal(680, 70, n).clip(300, 850).round(0),
})
df['RepaymentHistory'] = np.where(df['CreditScore'] > 650, 'Good',
                                  np.where(df['CreditScore'] > 550, 'Fair', 'Poor'))

plt.figure(figsize=(7, 5))
plt.hist(df['LoanAmount'], bins=25, color='steelblue', edgecolor='black')
plt.title("Distribution of Loan Amounts (Histogram)")
plt.xlabel("Loan Amount ($)"); plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
sns.boxplot(data=df, x='RepaymentHistory', y='LoanAmount',
            order=['Poor', 'Fair', 'Good'], hue='RepaymentHistory',
            palette='Set2', legend=False)
plt.title("Loan Amount by Repayment History (Box Plot)")
plt.xlabel("Repayment History"); plt.ylabel("Loan Amount ($)")
plt.tight_layout()
plt.show()

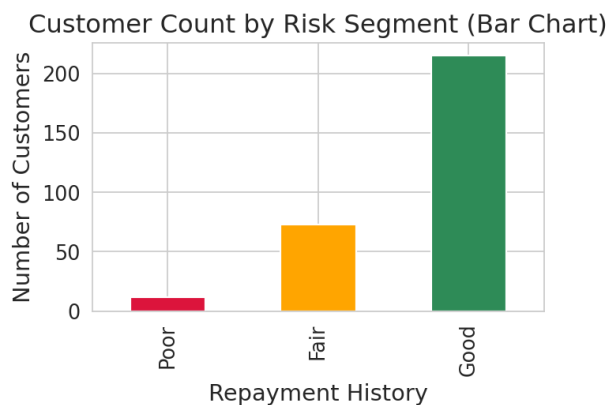
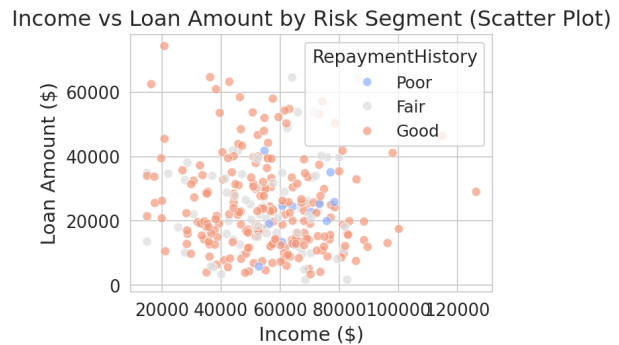
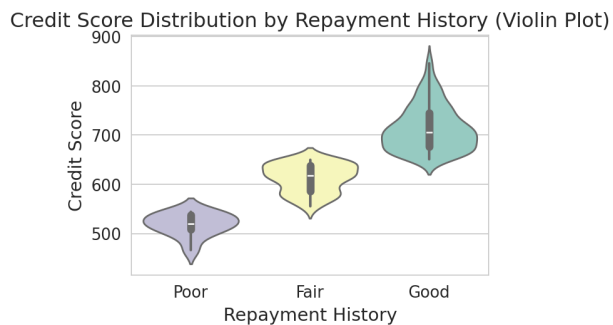
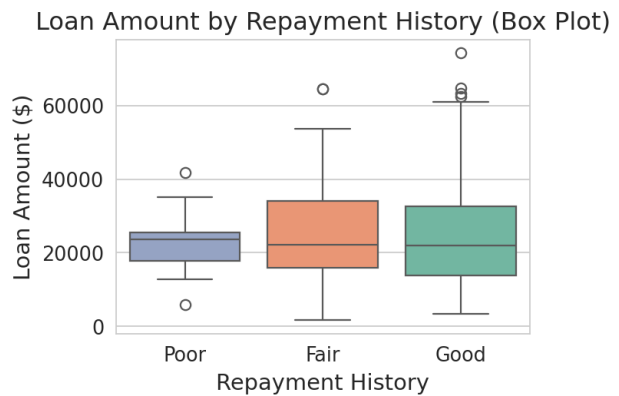
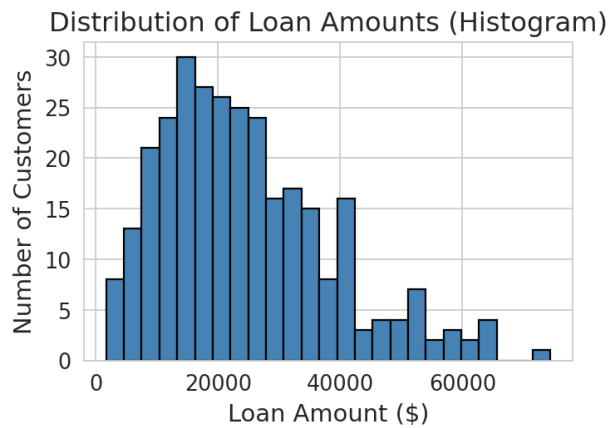
plt.figure(figsize=(7, 5))
sns.violinplot(data=df, x='RepaymentHistory', y='CreditScore',
               order=['Poor', 'Fair', 'Good'], hue='RepaymentHistory',
               palette='Set3', legend=False)
plt.title("Credit Score Distribution by Repayment History (Violin Plot)")
plt.xlabel("Repayment History"); plt.ylabel("Credit Score")
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
sns.scatterplot(data=df, x='Income', y='LoanAmount', hue='RepaymentHistory',
                hue_order=['Poor', 'Fair', 'Good'], palette='coolwarm', alpha=0.7)
plt.title("Income vs Loan Amount by Risk Segment (Scatter Plot)")
plt.xlabel("Income ($)"); plt.ylabel("Loan Amount ($)")
plt.tight_layout()
plt.show()

seg_counts = df['RepaymentHistory'].value_counts().reindex(['Poor', 'Fair', 'Good'])
plt.figure(figsize=(6, 5))
seg_counts.plot(kind='bar', color=['crimson', 'orange', 'seagreen'])
plt.title("Customer Count by Risk Segment (Bar Chart)")
plt.xlabel("Repayment History"); plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

print("Evaluation: see accompanying write-up for visualization justification.")
```

### Output



R

## Code

```
library(ggplot2)
library(dplyr)

set.seed(1)

n <- 300
df <- data.frame(
  CustomerID = 1:n,
  LoanAmount = round(rgamma(n, shape=3, scale=8000), 0),
  Income = round(pmax(rnorm(n, 55000, 18000), 15000), 0),
  CreditScore = round(pmin(pmax(rnorm(n, 680, 70), 300), 850), 0)
)
df$RepaymentHistory <- ifelse(df$CreditScore > 650, 'Good',
                              ifelse(df$CreditScore > 550, 'Fair', 'Poor'))
df$RepaymentHistory <- factor(df$RepaymentHistory, levels=c('Poor', 'Fair', 'Good'))

p1 <- ggplot(df, aes(x=LoanAmount)) +
  geom_histogram(bins=25, fill="steelblue", color="black") +
  labs(title="Distribution of Loan Amounts (Histogram)", x="Loan Amount ($)", y="Number of Customers") +
  theme_minimal()
print(p1)
p2 <- ggplot(df, aes(x=RepaymentHistory, y=LoanAmount, fill=RepaymentHistory)) +
```

```

geom_boxplot(show.legend=FALSE) +
labs(title="Loan Amount by Repayment History (Box Plot)", x="Repayment History", y="Loan Amount ($)") +
theme_minimal()
print(p2)

p3 <- ggplot(df, aes(x=RepaymentHistory, y=CreditScore, fill=RepaymentHistory)) +
geom_violin(show.legend=FALSE) +
labs(title="Credit Score Distribution by Repayment History (Violin Plot)",
x="Repayment History", y="Credit Score") +
theme_minimal()
print(p3)

p4 <- ggplot(df, aes(x=Income, y=LoanAmount, color=RepaymentHistory)) +
geom_point(alpha=0.7) +
scale_color_manual(values=c(Poor="firebrick", Fair="orange", Good="seagreen")) +
labs(title="Income vs Loan Amount by Risk Segment (Scatter Plot)", x="Income ($)", y="Loan Amount ($)") +
theme_minimal()
print(p4)

seg_counts <- df %>% group_by(RepaymentHistory) %>% summarise(Count=n())
p5 <- ggplot(seg_counts, aes(x=RepaymentHistory, y=Count, fill=RepaymentHistory)) +
geom_col(show.legend=FALSE) +
scale_fill_manual(values=c(Poor="firebrick", Fair="orange", Good="seagreen")) +
labs(title="Customer Count by Risk Segment (Bar Chart)", x="Repayment History", y="Number of Customers") +
theme_minimal()
print(p5)

```

## Output

